About 2,600 results for “review wildfire spread prediction”

Fields of Study | Date Range | Has PDF | Author | Journals & Conferences | More Filters | Filters | Sort by Relevance

Experimental and numerical study on data-driven prediction for wildfire spread incorporating adaptive observation error adjustment

Zheng Wang · Xingdong Li · Mengxia Zha · Jie Ji · Environmental Science, Engineering ·

Fire safety journal · 1 July 2024

2 · Publisher · Save · Cite

Application of Explainable Artificial Intelligence in Predicting Wildfire Spread: An ASPP-Enabled CNN Approach

Mohammad Marjani · M. Mahdianpari · Seyed Ali Ahmadi · Emadoddin Hemmati · F. Mohammadimanesh · Mohammad Saadi Mesgari

Environmental Science, Computer Science · IEEE Geoscience and Remote Sensing Letters · 2024

TLDR These findings indicate that larger dilation rates (DRs) can extract more meaningful features from the input data and contribute to the development of more transparent and accurate models for **wildfire spread prediction**, which could have significant implications for forest management and **wildfire** prevention strategies. Expand

17 · IEEE · Save · Cite

Rapid prediction of wildfire spread using ensemble Kalman filter and polyline simplification

Seungmin Yoo · Junho Song · Environmental Science, Engineering ·

Environmental Modelling & Software · 2023

4 · View Paper · Save · Cite

Wildfire Risk Prediction: A Review

Zhengsen Xu · Jonathan Li · Linlin Xu · Environmental Science, Computer Science · arXiv.org · 2024

TLDR The need for more e ff ective deep learning time series forecasting algorithms, the utilization of three-dimensional data including ground and trunk fuel, extraction of more accurate historical fire point data, and improved model evaluation metrics is emphasized. Expand

6 · Publisher · Save · Cite

Applying Probability Theory for the Quality Assessment of a Wildfire Spread Prediction Framework Based on Genetic Algorithms

A. Cencerrado · A. Cortés · T. Margalef · Computer Science, Environmental Science ·

TheScientificWorldJournal · 25 December 2013

TLDR A two-stage **prediction** process is relied on to enhance the quality of traditional **predictions**, taking advantage of parallel computing, based on an adjustment stage carried out by a well-known evolutionary technique: Genetic Algorithms. Expand

6 · [PDF] · PDF · Save · Cite

Wildfire spreading prediction using multimodal data and deep neural network approach

Dmitrii G. Shadrin · S. Illarionova · +5 authors · Evgeny Burnaev

Environmental Science, Computer Science · Scientific Reports · 31 January 2024

TLDR A neural network model based on the MA-Net architecture is trained to predict **wildfire spread** based on environmental and climate data, taking into account spatial distribution features, and shows promise for further transfer and adaptation to other regions. Expand

42 PDF PubMed Save Cite

Advancements in Wildfire Prediction and Detection: A Systematic Review

Vinothkumar Kolluru Advaitha Naidu Chintakunta Yudhisthir Nuthakki Sonika Koganti

Environmental Science, Computer Science · Social Science Research Network · 2024

TLDR The **review** underscores the importance of interdisciplinary collaboration and stakeholder engagement in addressing socio-ecological challenges associated with **wildfire**s, and highlights the need for continued innovation, data standardization, and knowledge exchange to advance **wildfire prediction** and detection capabilities. Expand

3 PDF Publisher Save Cite

Rapid prediction of wildfire spread using ensemble Kalman filter and polyline simplification

Seungmin Yoo Junho Song Environmental Science · Environmental Modelling & Software · 1 December 2022

9 Publisher Save Cite

Advancements in Wildfire Detection and Prediction: An In-Depth Review

Reem Salman Ali Karouni Elias Rachid Nizar Hamadeh

Environmental Science, Computer Science · International Journal of Innovative Technology… · 30 January 2024

TLDR Thirty articles published between 2004 and 2023 were **review**ed, summarizing the technologies utilized in forest fire **prediction** along with comprehensive analysis (surveys) of their techniques employed for this application. Expand

Publisher Save Cite

Robust Prediction of Wildfire Spread in Australia

Michael Palk Katharina Knappmann Stefan Voss Raka Jovanovic Environmental Science · IEEE Access · 2025

Wildfires can have devastating effects on urban infrastructure and natural ecosystems, making **wildfire** management an important, but yet complex and difficult task. The systematic collection of data,… Expand

IEEE Save Cite

← … 4 5 6 7 … →

Stay Connected With Semantic Scholar

Your E-mail Address Sign Up

What Is Semantic Scholar?

Semantic Scholar is a free, AI-powered research tool for scientific literature, based at Ai2.

Learn More

About
About Us
Publishers
Blog
Ai2 Careers

Product
Product Overview
Semantic Reader
Scholar's Hub
Beta Program
Release Notes

API
API Overview
API Tutorials
API Documentation
API Gallery

Research
Publications
Research Careers
Resources

Help
FAQ
Librarians
Tutorials
Contact

Proudly built by Ai2
Collaborators & Attributions • Terms of Service • Privacy Policy • API License Agreement